

Day 12 Review Sheet · Days 5-10

MathCraft · Grade 6 Mathematics · Professor Xavier Honablue, M.Ed.

Name: _____ Date: _____ Hour: _____

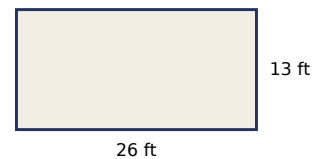
Covers: rectangle area & two-digit multiplication (Day 5) · missing sides (Day 6) · compound rectangles (Day 7) · parallelograms (Days 8-9), including extra practice spotting distractor slanted sides and diagonals · multiplying decimals & fractions (Day 10). **Standards:** 6.G.A.1, 6.NS.B.2, 6.NS.B.3, with prerequisite 5.NF.B.4.

Directions: Show ALL work by hand in your math notebook (graph paper is best). Label every answer with units. Figures are not drawn to scale. **Watch out:** some parallelograms show extra measurements — only base $\times$ height (the perpendicular distance) gives the area.

Section A · Area Is Multiplication (Day 5)

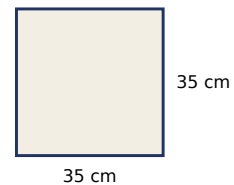
1. A rectangle is 26 ft long and 13 ft wide. Find its area.

Answer: _____



2. A square tile has sides of 35 cm. What is its area?

Answer: _____

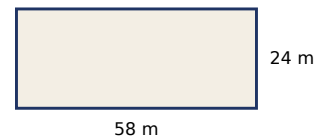


3. A classroom rug is 14 ft by 8 ft. How many 1-foot-by-1-foot squares would it take to cover the rug exactly?

Answer: _____

4. A community garden is 58 m long and 24 m wide. Find the area.

Answer: _____



5. **Find the mistake.** Jordan found the area of a 17 by 14 rectangle like this: $17 \times 10 = 170$, then $170 + 4 = 174$. What went wrong? What is the correct area?

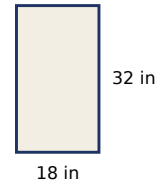
Answer: _____

6. Which rectangle has the greater area: 22 by 18 or 20 by 20? By how much?

Answer: _____

7. A poster is 18 in by 32 in. Find its area, and explain what the unit "square inches" means.

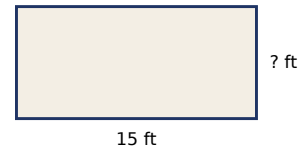
Answer: _____



Section B · Area Unlocks the Missing Side (Day 6)

8. A rectangle has an area of 105 sq ft. Its length is 15 ft. Find the width.

Answer: _____



9. A rectangle has an area of 437 sq m and a width of 19 m. Find the length.

Answer: _____

10. A square has an area of 169 sq in. How long is each side?

Answer: _____

11. **Square vs. rectangle repair.** Maya says, "Every rectangle is a square." Is she right? Explain and give an example.

Answer: _____

12. A rectangle has an area of 288 sq in and one side of 16 in. Find the other side, then find the perimeter.

Answer: _____

13. A square has an area of 64 sq cm. What is its perimeter?

Answer: _____

14. A rectangular gym floor has an area of 540 sq ft and is 27 ft long. How wide is it?

Answer: _____

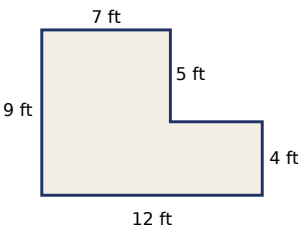
15. Two rectangles each have an area of 72 sq units. One has a side of 6; the other has a side of 8. Find each missing side. Which rectangle has the larger perimeter?

Answer: _____

Section C · Area of Compound Rectangles (Day 7)

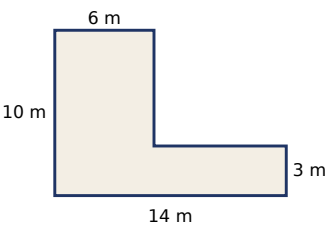
16. Find the area of the L-shaped floor by splitting it into two rectangles. (All angles are right angles.)

Answer: _____



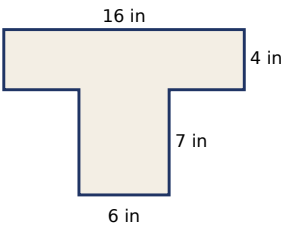
17. Find the area of this L-shape TWO ways: (a) split and add, (b) enclose and subtract. Show both give the same answer.

Answer: _____



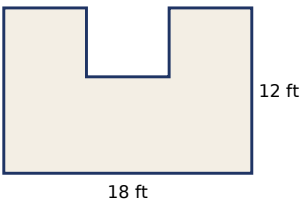
18. Find the area of the T-shape. The top bar is 16 in wide and 4 in tall. The stem is 6 in wide and 7 in tall.

Answer: _____



19. An 18 ft by 12 ft rectangle has a 6 ft wide, 5 ft deep notch cut out of the top middle. Find the area of the shape that remains.

Answer: _____

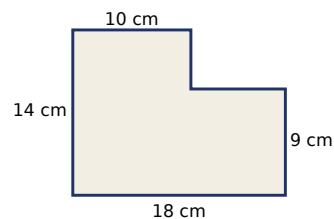


20. A bedroom is 18 ft by 14 ft. A 5 ft by 4 ft closet in the corner will NOT get carpet. Carpet costs \$4 per square foot. How much will the carpet cost?

Answer: _____

21. The two unlabeled side lengths are missing. Find them, then find the total area.

Answer: _____

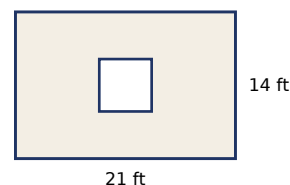


22. Find the mistake. For the floor in Question 16 (12 by 9 with a 5 by 5 notch), Deshawn wrote $12 \times 4 + 7 \times 9 = 48 + 63 = 111$ sq ft. What went wrong?

Answer: _____

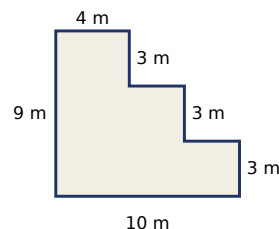
23. A rectangular patio is 21 ft by 14 ft. A 5 ft by 5 ft square flower bed in the middle will not be paved. How many square feet will be paved?

Answer: _____



24. A staircase shape is made of three stacked rectangles, each 3 m tall. Their widths are 10 m, 7 m, and 4 m. Find the total area.

Answer: _____



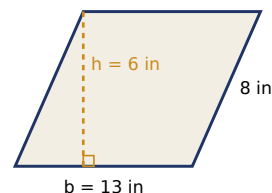
25. A T-shaped stage has a top section 24 ft by 5 ft and a stem 6 ft by 8 ft. It will be covered with square tiles that are 2 ft by 2 ft. How many tiles are needed?

Answer: _____

Section D · Area of a Parallelogram (Days 8-9)

26. Find the area of the parallelogram. (The dashed line is the height.)

Answer: _____



27. A parallelogram has a base of 17 cm and a height of 6 cm. Find its area.

Answer: _____

28. A parallelogram has an area of 144 sq in and a base of 16 in. Find its height.

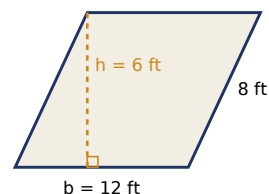
Answer: _____

29. A parallelogram has an area of 112 sq m and a height of 7 m. Find its base.

Answer: _____

30. Find the mistake. A parallelogram has base 12 ft, slanted side 8 ft, and height 6 ft. A student said the area is $12 \times 8 = 96$ sq ft. Explain the error and find the correct area.

Answer: _____



31. Explain why the area of a parallelogram is base $\times$ height. Use the idea of cutting and sliding.

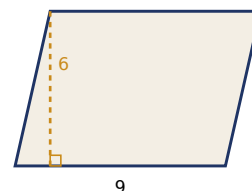
Answer: _____

32. A parallelogram-shaped garden has a base of 24 ft and a height of 11 ft. A 4 ft by 4 ft square pond sits inside it. How much area is left for plants?

Answer: _____

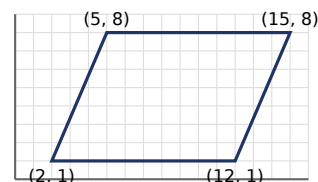
33. Two parallelograms both have a base of 9 cm and a height of 6 cm, but one leans much farther than the other. Compare their areas.

Answer: _____



34. On a coordinate grid, a parallelogram has vertices $(2, 1)$, $(12, 1)$, $(15, 8)$, and $(5, 8)$. Find its area.

Answer: _____

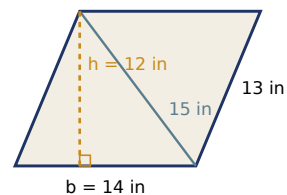


35. A rectangle is 11 in by 7 in. A parallelogram has a base of 11 in, a slanted side of 7 in, and a height of 5 in. Which has more area? How much more?

Answer: _____

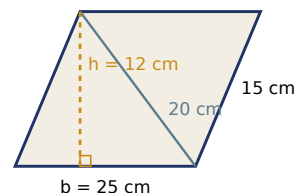
36. Distractor alert. This parallelogram shows FOUR measurements: the base, a slanted side, a diagonal, and the height. Find its area.

Answer: _____



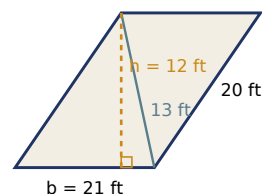
37. A parallelogram has a base of 25 cm, a slanted side of 15 cm, and a diagonal of 20 cm. Its height is 12 cm. Find its area.

Answer: _____



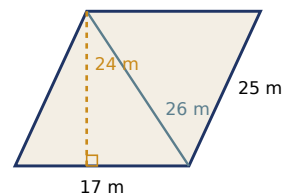
38. Find the mistake. A parallelogram has a base of 21 ft, a slanted side of 20 ft, a diagonal of 13 ft, and a height of 12 ft. Kiara said, "The diagonal goes up from the base, so it's the height," and wrote $21 \times 13 = 273$ sq ft. Explain her error and find the correct area.

Answer: _____



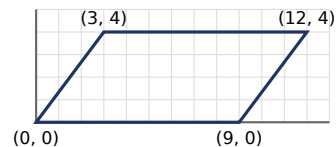
39. Which one is the height? A parallelogram is labeled 17 m, 25 m, 26 m, and 24 m. The 17 m side is the base. The 25 m segment is a side, the 26 m segment is a diagonal, and the 24 m segment makes a right angle with the base. Which is the height? Find the area.

Answer: _____



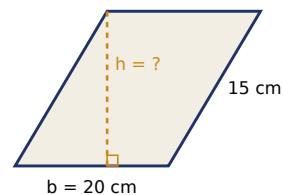
40. A parallelogram has vertices (0, 0), (9, 0), (12, 4), and (3, 4). Ty says, "The slanted side from (0, 0) to (3, 4) is 5 units long, so the height is 5." Is Ty right? Find the area.

Answer: _____



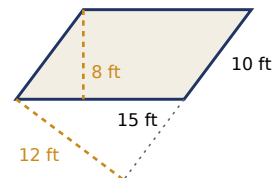
41. A parallelogram has an area of 180 sq cm and a base of 20 cm. Its slanted side is 15 cm. Find its height.

Answer: _____



42. Two bases, two heights. This parallelogram has sides 15 ft and 10 ft. The height to the 15 ft side is 8 ft, and the height to the 10 ft side is 12 ft. Sam wrote $15 \times 12 = 180$ sq ft. What went wrong? Find the area.

Answer: _____



Section E · Skill Breakout: Multiplying Decimals (Day 10)

43. Multiply: 0.7×0.3

Answer: _____

44. Multiply: 2.4×1.5

Answer: _____

45. Multiply: 3.06×5

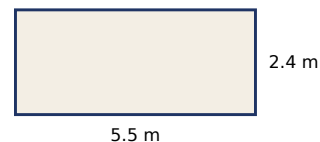
Answer: _____

46. Multiply: 7.5×0.04

Answer: _____

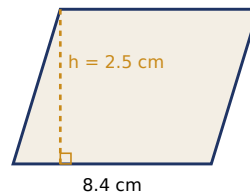
47. A rectangular rug is 5.5 m by 2.4 m. Find its area.

Answer: _____



48. A parallelogram has a base of 8.4 cm and a height of 2.5 cm. Find its area.

Answer: _____



49. Find the mistake. Priya says $0.4 \times 0.2 = 0.8$. Explain her error and give the correct product.

Answer: _____

Section F · Skill Breakout: Multiplying Fractions (Day 10)

50. Multiply: $\frac{3}{4} \times \frac{2}{9}$. Simplify.

Answer: _____

51. Find $\frac{2}{3}$ of 24.

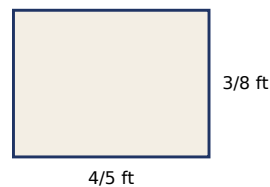
Answer: _____

52. Multiply: $1\frac{1}{3} \times 2\frac{1}{4}$

Answer: _____

53. A rectangle is $\frac{4}{5}$ ft long and $\frac{3}{8}$ ft wide. Find its area.

Answer: _____



54. Same function, different notation. Compute 0.5×44 and $\frac{1}{2} \times 44$. Then compute 0.25×64 and $\frac{1}{4} \times 64$. What do you notice? Explain.

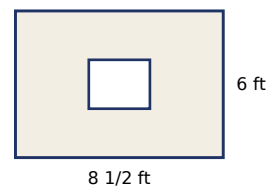
Answer: _____

55. A parallelogram has a base of $6\frac{1}{2}$ in and a height of 6 in. Find its area.

Answer: _____

56. Challenge. A rectangular yard is $8\frac{1}{2}$ ft by 6 ft. A shed that is 2.5 ft by 2 ft sits in the yard. How many square feet of yard are NOT covered by the shed?

Answer: _____



Answer Key & In-Depth Solutions

Day 12 Review · Days 5–10 · Teacher / family reference — try every problem before checking!

Section A · Area Is Multiplication (Day 5)

1. A rectangle is 26 ft long and 13 ft wide. Find its area.

- Step 1: Area of a rectangle = length $\times$ width, so we need 26×13 .
- Step 2: Break both factors apart (area model): $26 = 20 + 6$ and $13 = 10 + 3$.
- Step 3: Multiply every part: $20 \times 10 = 200$, $20 \times 3 = 60$, $6 \times 10 = 60$, $6 \times 3 = 18$.
- Step 4: Add the partial products: $200 + 60 + 60 + 18 = 338$.
- Step 5: Units: feet $\times$ feet = square feet. Area = **338 sq ft**.

✓ **Answer: 338 sq ft**

2. A square tile has sides of 35 cm. What is its area?

- Step 1: A square is a rectangle with all sides equal, so Area = side $\times$ side = 35×35 .
- Step 2: $35 \times 35 = (30 + 5)(30 + 5) = 900 + 150 + 150 + 25 = 1,225$.
- Step 3: Area = **1,225 sq cm**.

✓ **Answer: 1,225 sq cm**

3. A classroom rug is 14 ft by 8 ft. How many 1-foot-by-1-foot squares would it take to cover the rug exactly?

- Step 1: Each 1 ft $\times$ 1 ft square covers exactly 1 square foot, so the number of squares equals the area.
- Step 2: Think of rows: there are 8 rows with 14 squares in each row.
- Step 3: $14 \times 8 = (10 \times 8) + (4 \times 8) = 80 + 32 = 112$.
- Step 4: It takes **112 squares** (112 sq ft).

✓ **Answer: 112 squares**

4. A community garden is 58 m long and 24 m wide. Find the area.

- Step 1: Area = 58×24 .
- Step 2: Area model: $58 = 50 + 8$ and $24 = 20 + 4$.
- Step 3: $50 \times 20 = 1,000$; $50 \times 4 = 200$; $8 \times 20 = 160$; $8 \times 4 = 32$.
- Step 4: $1,000 + 200 + 160 + 32 = 1,392$.
- Step 5: Estimate check: $60 \times 24 = 1,440$, so 1,392 is reasonable.
- Step 6: Area = **1,392 sq m**.

✓ **Answer: 1,392 sq m**

5. Find the mistake. Jordan found the area of a 17 by 14 rectangle like this: $17 \times 10 = 170$, then $170 + 4 = 174$. What went wrong? What is the correct area?

- Step 1: $14 = 10 + 4$, so Jordan must multiply 17 by BOTH parts.
- Step 2: Jordan did $17 \times 10 = 170$ correctly, but then only added 4 instead of multiplying 17×4 .
- Step 3: The missing partial product is $17 \times 4 = 68$.
- Step 4: Correct area: $170 + 68 = 238$.
- Step 5: Area = **238 square units**. (Lesson: every part of one factor gets multiplied by every part of the other.)

✓ **Answer: 238 square units**

6. Which rectangle has the greater area: 22 by 18 or 20 by 20? By how much?

- Step 1: 22×18 : $(20 + 2)(10 + 8) = 200 + 160 + 20 + 16 = 396$.
- Step 2: $20 \times 20 = 400$.
- Step 3: $400 - 396 = 4$.
- Step 4: The **20 by 20 square is greater by 4 square units**. Both have the same perimeter (80), but the square shape holds a little more area.

✓ **Answer: 20 × 20 is greater, by 4 square units**

7. A poster is 18 in by 32 in. Find its area, and explain what the unit "square inches" means.

- Step 1: Area = 18×32 .
- Step 2: $18 \times 32 = (10 + 8)(30 + 2) = 300 + 20 + 240 + 16 = 576$.
- Step 3: "Square inches" means we are counting how many 1-inch-by-1-inch squares cover the poster.
- Step 4: Area = **576 sq in**: 576 tiny 1-inch squares would cover it exactly.

✓ **Answer: 576 sq in**

Section B · Area Unlocks the Missing Side (Day 6)

8. A rectangle has an area of 105 sq ft. Its length is 15 ft. Find the width.

- Step 1: Area = length $\times$ width, so $105 = 15 \times \text{width}$.
- Step 2: Run area in reverse: width = area $\div$ length = $105 \div 15$.
- Step 3: $15 \times 7 = 105$, so $105 \div 15 = 7$.
- Step 4: Width = **7 ft**. Check: $15 \times 7 = 105$. ✓

✓ **Answer: 7 ft**

9. A rectangle has an area of 437 sq m and a width of 19 m. Find the length.

- Step 1: Length = $437 \div 19$.
- Step 2: Estimate: $19 \times 20 = 380$, which leaves $437 - 380 = 57$.
- Step 3: $19 \times 3 = 57$, so we need $20 + 3 = 23$ groups of 19.
- Step 4: Length = **23 m**. Check: $19 \times 23 = 380 + 57 = 437$. ✓

✓ **Answer: 23 m**

10. A square has an area of 169 sq in. How long is each side?

- Step 1: In a square, both sides are the same number, so we need a number times itself that equals 169.
- Step 2: Try: $12 \times 12 = 144$ (too small), $13 \times 13 = 169$. ✓
- Step 3: Each side = **13 in**.

✓ **Answer: 13 in**

11. Square vs. rectangle repair. Maya says, "Every rectangle is a square." Is she right? Explain and give an example.

- Step 1: A rectangle is a four-sided shape with four right angles.
- Step 2: A square is a rectangle that ALSO has all four sides equal.
- Step 3: So a square is a special kind of rectangle, but a rectangle does not have to have equal sides.
- Step 4: Counterexample: a 3 by 8 rectangle has four right angles but its sides are not all equal, so it is not a square.
- Step 5: Maya has it backwards: **every square is a rectangle, but not every rectangle is a square.**

✓ **Answer: No — every square is a rectangle, but not every rectangle is a square.**

12. A rectangle has an area of 288 sq in and one side of 16 in. Find the other side, then find the perimeter.

- Step 1: Other side = $288 \div 16$.
- Step 2: $16 \times 10 = 160 \rightarrow 288 - 160 = 128$. $16 \times 8 = 128$. So $10 + 8 = 18$.
- Step 3: The other side is 18 in. Check: $16 \times 18 = 288$. ✓
- Step 4: Perimeter = $2 \times (16 + 18) = 2 \times 34 = 68$.
- Step 5: Other side = **18 in**; perimeter = **68 in**. (Area is in square inches, perimeter is in plain inches.)

✓ **Answer: 18 in; perimeter 68 in**

13. A square has an area of 64 sq cm. What is its perimeter?

- Step 1: Find the side first: which number times itself is 64? $8 \times 8 = 64$, so the side is 8 cm.
- Step 2: A square has 4 equal sides: perimeter = $4 \times 8 = 32$.
- Step 3: Perimeter = **32 cm**.

✓ **Answer: 32 cm**

14. A rectangular gym floor has an area of 540 sq ft and is 27 ft long. How wide is it?

- Step 1: Width = $540 \div 27$.
- Step 2: Think: $27 \times 2 = 54$, so $27 \times 20 = 540$.
- Step 3: Width = **20 ft**. Check: $27 \times 20 = 540$. ✓

✓ **Answer: 20 ft**

15. Two rectangles each have an area of 72 sq units. One has a side of 6; the other has a side of 8. Find each missing side. Which rectangle has the larger perimeter?

- Step 1: Rectangle 1: $72 \div 6 = 12$, so it is 6 by 12.
- Step 2: Rectangle 2: $72 \div 8 = 9$, so it is 8 by 9.
- Step 3: Perimeter 1: $2 \times (6 + 12) = 36$. Perimeter 2: $2 \times (8 + 9) = 34$.
- Step 4: Missing sides: **12 and 9**. The **6 by 12** rectangle has the larger perimeter ($36 > 34$), even though the areas are equal.

✓ **Answer: 12 and 9; the 6 × 12 rectangle (36 vs. 34)**

Section C · Area of Compound Rectangles (Day 7)

16. Find the area of the L-shaped floor by splitting it into two rectangles. (All angles are right angles.)

- Step 1: Find any hidden sides: the notch is $12 - 7 = 5$ ft wide and $9 - 4 = 5$ ft tall.
- Step 2: Split with a vertical cut at 7 ft. Left rectangle: $7 \times 9 = 63$.
- Step 3: Right rectangle: 5 ft wide ($12 - 7$) and 4 ft tall $\rightarrow 5 \times 4 = 20$.
- Step 4: Add: $63 + 20 = 83$.
- Step 5: Area = **83 sq ft**.

✓ **Answer: 83 sq ft**

17. Find the area of this L-shape TWO ways: (a) split and add, (b) enclose and subtract. Show both give the same answer.

- Step 1: (a) Split and add: cut horizontally at height 3. Bottom rectangle: $14 \times 3 = 42$. Top rectangle: 6 wide $\times$ ($10 - 3$) = $6 \times 7 = 42$. Total $42 + 42 = 84$.
- Step 2: (b) Enclose and subtract: the enclosing rectangle is $14 \times 10 = 140$.
- Step 3: The missing (empty) piece is ($14 - 6$) wide by ($10 - 3$) tall = $8 \times 7 = 56$.
- Step 4: $140 - 56 = 84$.
- Step 5: Both methods give **84 sq m**. ✓

✓ **Answer: 84 sq m**

18. Find the area of the T-shape. The top bar is 16 in wide and 4 in tall. The stem is 6 in wide and 7 in tall.

- Step 1: Split into the top bar and the stem.
- Step 2: Top bar: $16 \times 4 = 64$.
- Step 3: Stem: $6 \times 7 = 42$.
- Step 4: Add: $64 + 42 = 106$.
- Step 5: Check by subtracting: enclosing rectangle $16 \times 11 = 176$. Each empty corner is $5 \times 7 = 35$ (because $(16 - 6) \div 2 = 5$). $176 - 35 - 35 = 106$. ✓
- Step 6: Area = **106 sq in**.

✓ **Answer: 106 sq in**

19. An 18 ft by 12 ft rectangle has a 6 ft wide, 5 ft deep notch cut out of the top middle. Find the area of the shape that remains.

- Step 1: Enclose and subtract is fastest here.
- Step 2: Whole rectangle: $18 \times 12 = 216$.
- Step 3: Notch (the missing piece): $6 \times 5 = 30$.
- Step 4: $216 - 30 = 186$.
- Step 5: Area = **186 sq ft**.

✓ **Answer: 186 sq ft**

20. A bedroom is 18 ft by 14 ft. A 5 ft by 4 ft closet in the corner will NOT get carpet. Carpet costs \$4 per square foot. How much will the carpet cost?

- Step 1: Room area: $18 \times 14 = 252$ sq ft ($18 \times 10 = 180$, $18 \times 4 = 72$).
- Step 2: Closet area: $5 \times 4 = 20$ sq ft.
- Step 3: Carpeted area: $252 - 20 = 232$ sq ft.
- Step 4: Cost: $232 \times \$4 = \928 ($200 \times 4 = 800$, $30 \times 4 = 120$, $2 \times 4 = 8$; $800 + 120 + 8 = 928$).
- Step 5: The carpet costs **\$928**.

✓ **Answer: \$928**

21. The two unlabeled side lengths are missing. Find them, then find the total area.

- Step 1: Missing horizontal side: the bottom is 18 and the top is 10, so $18 - 10 = 8$ cm.
- Step 2: Missing vertical side: the left is 14 and the right is 9, so $14 - 9 = 5$ cm.
- Step 3: Split horizontally at height 9: bottom $18 \times 9 = 162$; top $10 \times 5 = 50$.
- Step 4: $162 + 50 = 212$.
- Step 5: Check: $18 \times 14 = 252$; empty corner $8 \times 5 = 40$; $252 - 40 = 212$. ✓
- Step 6: Missing sides: **8 cm and 5 cm**; area = **212 sq cm**.

✓ **Answer: 8 cm and 5 cm; area 212 sq cm**

22. Find the mistake. For the floor in Question 16 (12 by 9 with a 5 by 5 notch), Deshawn wrote $12 \times 4 + 7 \times 9 = 48 + 63 = 111$ sq ft. What went wrong?

- Step 1: The 12×4 rectangle is the bottom strip. The 7×9 rectangle is the whole left side.
- Step 2: These two rectangles OVERLAP in the bottom-left corner: a $7 \times 4 = 28$ sq ft region got counted twice.
- Step 3: Fix: $111 - 28 = 83$, OR use non-overlapping pieces: $7 \times 9 + 5 \times 4 = 63 + 20 = 83$.
- Step 4: Rule: when you split a shape, the pieces must not overlap. Correct area = **83 sq ft**.

✓ **Answer: He counted the 7×4 overlap twice; correct area is 83 sq ft**

23. A rectangular patio is 21 ft by 14 ft. A 5 ft by 5 ft square flower bed in the middle will not be paved. How many square feet will be paved?

- Step 1: Patio area: $21 \times 14 = 294$ ($21 \times 10 = 210$, $21 \times 4 = 84$).
- Step 2: Flower bed: $5 \times 5 = 25$.
- Step 3: $294 - 25 = 269$.
- Step 4: Paved area = **269 sq ft**.

✓ **Answer: 269 sq ft**

24. A staircase shape is made of three stacked rectangles, each 3 m tall. Their widths are 10 m, 7 m, and 4 m. Find the total area.

- Step 1: Bottom step: $10 \times 3 = 30$.
- Step 2: Middle step: $7 \times 3 = 21$.
- Step 3: Top step: $4 \times 3 = 12$.
- Step 4: Add: $30 + 21 + 12 = 63$.
- Step 5: Check: enclosing rectangle $10 \times 9 = 90$; empty pieces $6 \times 3 = 18$ (top row) and $3 \times 3 = 9$ (middle row); $90 - 18 - 9 = 63$. ✓
- Step 6: Area = **63 sq m**.

✓ **Answer: 63 sq m**

25. A T-shaped stage has a top section 24 ft by 5 ft and a stem 6 ft by 8 ft. It will be covered with square tiles that are 2 ft by 2 ft. How many tiles are needed?

- Step 1: Top section: $24 \times 5 = 120$ sq ft. Stem: $6 \times 8 = 48$ sq ft.
- Step 2: Total stage area: $120 + 48 = 168$ sq ft.
- Step 3: Each tile covers $2 \times 2 = 4$ sq ft (NOT 2 sq ft!).
- Step 4: Number of tiles: $168 \div 4 = 42$.
- Step 5: They need **42 tiles**.

✓ **Answer: 42 tiles**

Section D · Area of a Parallelogram (Days 8-9)

26. Find the area of the parallelogram. (The dashed line is the height.)

- Step 1: Area of a parallelogram = base $\times$ height (the height is perpendicular to the base).
- Step 2: Base = 13 in; height = 6 in. The slanted side (8 in) is NOT the height — ignore it.
- Step 3: $13 \times 6 = 78$.
- Step 4: Area = **78 sq in**.

✓ **Answer: 78 sq in**

27. A parallelogram has a base of 17 cm and a height of 6 cm. Find its area.

- Step 1: $A = b \times h = 17 \times 6$.
- Step 2: $17 \times 6 = (10 \times 6) + (7 \times 6) = 60 + 42 = 102$.
- Step 3: Area = **102 sq cm**.

✓ **Answer: 102 sq cm**

28. A parallelogram has an area of 144 sq in and a base of 16 in. Find its height.

- Step 1: $A = b \times h$, so $144 = 16 \times h$.
- Step 2: $h = 144 \div 16$.
- Step 3: $16 \times 9 = 144$, so $h = 9$.
- Step 4: Height = **9 in**. Check: $16 \times 9 = 144$. ✓

✓ **Answer: 9 in**

29. A parallelogram has an area of 112 sq m and a height of 7 m. Find its base.

- Step 1: $112 = b \times 7$, so $b = 112 \div 7$.
- Step 2: $7 \times 16 = 112$ ($7 \times 10 = 70$, $7 \times 6 = 42$).
- Step 3: Base = **16 m**.

✓ **Answer: 16 m**

30. Find the mistake. A parallelogram has base 12 ft, slanted side 8 ft, and height 6 ft. A student said the area is $12 \times 8 = 96$ sq ft. Explain the error and find the correct area.

- Step 1: The student used base $\times$ SLANT. The formula needs base $\times$ HEIGHT.
- Step 2: The slanted side leans over, so it is longer than the straight-up height ($8 > 6$). Using it makes the area too big.
- Step 3: Picture it: cut the triangle off one end and slide it to the other — the new rectangle is 12 long and 6 tall, not 8 tall.
- Step 4: Correct area: $12 \times 6 = 72$.
- Step 5: Area = **72 sq ft**.

✓ **Answer: 72 sq ft**

31. Explain why the area of a parallelogram is base $\times$ height. Use the idea of cutting and sliding.

- Step 1: Draw the height from one top corner straight down to the base. This cuts off a right triangle.
- Step 2: Slide that triangle to the opposite end. It fits exactly into the gap.
- Step 3: The new shape is a rectangle. No area was added or removed — we only moved a piece.
- Step 4: The rectangle's length is the parallelogram's base, and its width is the parallelogram's height.
- Step 5: Rectangle area = length $\times$ width, so **parallelogram area = base $\times$ height**.

✓ **Answer: It rearranges into a rectangle with the same base and height.**

32. A parallelogram-shaped garden has a base of 24 ft and a height of 11 ft. A 4 ft by 4 ft square pond sits inside it. How much area is left for plants?

- Step 1: Garden: $24 \times 11 = 264$ sq ft.
- Step 2: Pond: $4 \times 4 = 16$ sq ft.
- Step 3: $264 - 16 = 248$.
- Step 4: Planting area = **248 sq ft**.

✓ **Answer: 248 sq ft**

33. Two parallelograms both have a base of 9 cm and a height of 6 cm, but one leans much farther than the other. Compare their areas.

- Step 1: Area depends only on base and height: $9 \times 6 = 54$ for each.
- Step 2: How far it leans changes the slanted side length, but not the height.
- Step 3: Both areas = **54 sq cm**. They are equal.

✓ **Answer: Both are 54 sq cm — equal.**

34. On a coordinate grid, a parallelogram has vertices (2, 1), (12, 1), (15, 8), and (5, 8). Find its area.

- Step 1: The bottom side runs from (2, 1) to (12, 1): $\text{base} = 12 - 2 = 10$ units.
- Step 2: The top side is at $y = 8$ and the base is at $y = 1$, so $\text{height} = 8 - 1 = 7$ units (count straight up, not along the slant).
- Step 3: $A = 10 \times 7 = 70$.
- Step 4: Area = **70 square units**.

✓ **Answer: 70 square units**

35. A rectangle is 11 in by 7 in. A parallelogram has a base of 11 in, a slanted side of 7 in, and a height of 5 in. Which has more area? How much more?

- Step 1: Rectangle: $11 \times 7 = 77$ sq in.
- Step 2: Parallelogram: $\text{base} \times \text{height} = 11 \times 5 = 55$ sq in (not 11×7 !).
- Step 3: $77 - 55 = 22$.
- Step 4: The **rectangle has 22 sq in more**, even though the side lengths match.

✓ **Answer: Rectangle, by 22 sq in**

36. Distractor alert. This parallelogram shows FOUR measurements: the base, a slanted side, a diagonal, and the height. Find its area.

- Step 1: Area of a parallelogram = $\text{base} \times \text{height}$. Only two of the four numbers belong in the formula.
- Step 2: Base = 14 in. The height is the dashed segment that meets the base at a right angle (the little square): 12 in.
- Step 3: The slanted side (13 in) leans, and the diagonal (15 in) cuts corner to corner — neither one is perpendicular to the base, so both are distractors.
- Step 4: $14 \times 12 = (14 \times 10) + (14 \times 2) = 140 + 28 = 168$.
- Step 5: Area = **168 sq in**.

✓ **Answer: 168 sq in**

37. A parallelogram has a base of 25 cm, a slanted side of 15 cm, and a diagonal of 20 cm. Its height is 12 cm. Find its area.

- Step 1: Circle the base (25 cm) and the height (12 cm). Cross out the slanted side (15 cm) and the diagonal (20 cm).
- Step 2: Why? The diagonal is the longest segment inside the shape, but it is tilted — it is not straight up from the base.
- Step 3: $A = b \times h = 25 \times 12$.
- Step 4: $25 \times 12 = (25 \times 10) + (25 \times 2) = 250 + 50 = 300$.
- Step 5: Area = **300 sq cm**. (Trap answers: $25 \times 15 = 375$ and $25 \times 20 = 500$ are both too big.)

✓ **Answer: 300 sq cm**

38. Find the mistake. A parallelogram has a base of 21 ft, a slanted side of 20 ft, a diagonal of 13 ft, and a height of 12 ft. Kiara said, "The diagonal goes up from the base, so it's the height," and wrote $21 \times 13 = 273$ sq ft. Explain her error and find the correct area.

- Step 1: Kiara used the **DIAGONAL**. The diagonal starts on the base, but it runs at a slant to the opposite corner.
- Step 2: The height must be perpendicular (a 90° angle) to the base. That is the dashed 12 ft segment.
- Step 3: A slanted segment is always longer than the straight-up height ($13 > 12$), so her answer is too big.
- Step 4: Correct area: $21 \times 12 = (20 \times 12) + (1 \times 12) = 240 + 12 = 252$.
- Step 5: Area = **252 sq ft**. (The 20 ft slanted side is a distractor too.)

✓ **Answer: 252 sq ft**

39. Which one is the height? A parallelogram is labeled 17 m, 25 m, 26 m, and 24 m. The 17 m side is the base. The 25 m segment is a side, the 26 m segment is a diagonal, and the 24 m segment makes a right angle with the base. Which is the height? Find the area.

- Step 1: The height is the segment that makes a **right angle** with the base: 24 m.
- Step 2: 25 m is a slanted side and 26 m is a diagonal — both lean, so neither is the height.
- Step 3: Notice: 24 is the **SMALLEST** of the three up-and-down measurements. The straight-up distance is always the shortest path.
- Step 4: $A = 17 \times 24 = (17 \times 20) + (17 \times 4) = 340 + 68 = 408$.
- Step 5: Height = 24 m; area = **408 sq m**.

✓ **Answer: Height is 24 m; area 408 sq m**

40. A parallelogram has vertices (0, 0), (9, 0), (12, 4), and (3, 4). Ty says, "The slanted side from (0, 0) to (3, 4) is 5 units long, so the height is 5." Is Ty right? Find the area.

- Step 1: The slanted side really is 5 units long (it goes 3 over and 4 up), but it is **NOT** the height.
- Step 2: Height = straight-up distance between the bottom ($y = 0$) and the top ($y = 4$): $4 - 0 = 4$ units.
- Step 3: Base: (0, 0) to (9, 0) $\rightarrow 9 - 0 = 9$ units.
- Step 4: $A = 9 \times 4 = 36$.
- Step 5: Ty is **not** right. Area = **36 square units** (not $9 \times 5 = 45$).

✓ **Answer: No — height is 4; area 36 square units**

41. A parallelogram has an area of 180 sq cm and a base of 20 cm. Its slanted side is 15 cm. Find its height.

- Step 1: Area = base $\times$ height, so $180 = 20 \times h$.
- Step 2: The slanted side (15 cm) is extra information — it does not go in the formula.
- Step 3: $h = 180 \div 20 = 9$.
- Step 4: Check: $20 \times 9 = 180$. ✓ And $9 < 15$, which makes sense: the height is shorter than the slanted side.
- Step 5: Height = **9 cm**.

✓ **Answer: 9 cm**

42. Two bases, two heights. This parallelogram has sides 15 ft and 10 ft. The height to the 15 ft side is 8 ft, and the height to the 10 ft side is 12 ft. Sam wrote $15 \times 12 = 180$ sq ft. What went wrong? Find the area.

- Step 1: Every base has its OWN height — the one that is perpendicular to that base.
- Step 2: Sam mixed them up: the 12 ft height goes with the 10 ft side, not the 15 ft side.
- Step 3: Pair 1: $15 \times 8 = 120$.
- Step 4: Pair 2 (check): $10 \times 12 = 120$. ✓ Both pairs give the same area.
- Step 5: Area = **120 sq ft**.

✓ **Answer: Mismatched base and height; area 120 sq ft**

Section E · Skill Breakout: Multiplying Decimals (Day 10)

43. Multiply: 0.7×0.3

- Step 1: Ignore the decimals and multiply the whole numbers: $7 \times 3 = 21$.
- Step 2: Count decimal places in the factors: 0.7 has 1, 0.3 has 1 → 2 total.
- Step 3: Place the decimal so the answer has 2 decimal places: 0.21.
- Step 4: Meaning: 7 tenths $\times$ 3 tenths = 21 hundredths. Answer: **0.21**.

✓ **Answer: 0.21**

44. Multiply: 2.4×1.5

- Step 1: Multiply as whole numbers: $24 \times 15 = 24 \times 10 + 24 \times 5 = 240 + 120 = 360$.
- Step 2: Decimal places: $1 + 1 = 2$, so $360 \rightarrow 3.60$.
- Step 3: $3.60 = 3.6$. Estimate check: 2.4×1.5 is about $2 \times 1.5 = 3$ (a bit more), close to 3.6. ✓
- Step 4: Answer: **3.6**.

✓ **Answer: 3.6**

45. Multiply: 3.06×5

- Step 1: $306 \times 5 = 1,530$.
- Step 2: Decimal places: 2 (from 3.06) + 0 = 2, so $1530 \rightarrow 15.30$.
- Step 3: Estimate: $3 \times 5 = 15$. ✓
- Step 4: Answer: **15.3**.

✓ **Answer: 15.3**

46. Multiply: 7.5×0.04

- Step 1: $75 \times 4 = 300$.
- Step 2: Decimal places: 1 (7.5) + 2 (0.04) = 3, so $300 \rightarrow 0.300$.
- Step 3: $0.300 = 0.3$. Estimate: $8 \times 0.04 = 0.32$, so about 0.3. ✓
- Step 4: Answer: **0.3**.

✓ **Answer: 0.3**

47. A rectangular rug is 5.5 m by 2.4 m. Find its area.

- Step 1: Area = 5.5×2.4 .
- Step 2: $55 \times 24 = 55 \times 20 + 55 \times 4 = 1,100 + 220 = 1,320$.
- Step 3: Decimal places: $1 + 1 = 2 \rightarrow 13.20$.
- Step 4: Estimate: 5.5×2.5 is about 14, close. ✓
- Step 5: Area = **13.2 sq m**.

✓ **Answer: 13.2 sq m**

48. A parallelogram has a base of 8.4 cm and a height of 2.5 cm. Find its area.

- Step 1: $A = b \times h = 8.4 \times 2.5$.
- Step 2: $84 \times 25 = 2,100$ (because $4 \times 25 = 100$, so $84 \times 25 = 21 \times 100$).
- Step 3: Decimal places: $1 + 1 = 2 \rightarrow 21.00$.
- Step 4: Area = **21 sq cm**.

✓ **Answer: 21 sq cm**

49. Find the mistake. Priya says $0.4 \times 0.2 = 0.8$. Explain her error and give the correct product.

- Step 1: $4 \times 2 = 8$ is correct, but Priya did not count decimal places.
- Step 2: 0.4 has 1 decimal place and 0.2 has 1, so the product needs 2: 0.08.
- Step 3: Reasoning check: 0.4×0.2 means "4 tenths OF 2 tenths." Taking part of a number smaller than 1 gives something even smaller — 0.8 is bigger than both factors, so it cannot be right.
- Step 4: Correct product: **0.08**.

✓ **Answer: 0.08**

Section F · Skill Breakout: Multiplying Fractions (Day 10)

50. Multiply: $\frac{3}{4} \times \frac{2}{9}$. Simplify.

- Step 1: Multiply numerators: $3 \times 2 = 6$. Multiply denominators: $4 \times 9 = 36$.
- Step 2: $\frac{6}{36}$ simplifies: divide top and bottom by 6 $\rightarrow \frac{1}{6}$.
- Step 3: Shortcut: cross-simplify first — 3 and 9 become 1 and 3; 2 and 4 become 1 and 2. Then $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$.
- Step 4: Answer: **$\frac{1}{6}$** .

✓ **Answer: $\frac{1}{6}$**

51. Find $\frac{2}{3}$ of 24.

- Step 1: "Of" means multiply: $\frac{2}{3} \times 24$.
- Step 2: Find $\frac{1}{3}$ of 24 first: $24 \div 3 = 8$.
- Step 3: Then $\frac{2}{3}$ is two of those: $2 \times 8 = 16$.
- Step 4: Answer: **16**.

✓ **Answer: 16**

52. Multiply: $1\frac{1}{3} \times 2\frac{1}{4}$

- Step 1: Change mixed numbers to improper fractions: $1\frac{1}{3} = \frac{4}{3}$ and $2\frac{1}{4} = \frac{9}{4}$.
- Step 2: $\frac{4}{3} \times \frac{9}{4} = \frac{36}{12}$.
- Step 3: $\frac{36}{12} = 3$.
- Step 4: Estimate: about $1.3 \times 2.25 \approx 3$. ✓ Answer: **3**.

✓ Answer: **3**

53. A rectangle is $\frac{4}{5}$ ft long and $\frac{3}{8}$ ft wide. Find its area.

- Step 1: Area = $\frac{4}{5} \times \frac{3}{8} = \frac{12}{40}$.
- Step 2: $\frac{12}{40} = \frac{3}{10}$ (divide top and bottom by 4).
- Step 3: This makes sense: the rectangle is smaller than 1 ft by 1 ft, so its area should be less than 1 sq ft.
- Step 4: Area = **$\frac{3}{10}$ sq ft**.

✓ Answer: **$\frac{3}{10}$ sq ft**

54. Same function, different notation. Compute 0.5×44 and $\frac{1}{2} \times 44$. Then compute 0.25×64 and $\frac{1}{4} \times 64$. What do you notice? Explain.

- Step 1: 0.5×44 : $5 \times 44 = 220$, one decimal place $\rightarrow 22.0 = 22$.
- Step 2: $\frac{1}{2} \times 44 = 44 \div 2 = 22$.
- Step 3: 0.25×64 : $25 \times 64 = 1,600$, two decimal places $\rightarrow 16.00 = 16$.
- Step 4: $\frac{1}{4} \times 64 = 64 \div 4 = 16$.
- Step 5: Because $0.5 = \frac{1}{2}$ and $0.25 = \frac{1}{4}$, each pair is the SAME multiplication written two ways.
- Step 6: Results: **$22 = 22$ and $16 = 16$** . Pick whichever notation makes the math easier.

✓ Answer: **$22 = 22$ and $16 = 16$ — the decimal and fraction name the same number.**

55. A parallelogram has a base of $6\frac{1}{2}$ in and a height of 6 in. Find its area.

- Step 1: $A = b \times h = 6\frac{1}{2} \times 6$.
- Step 2: Distribute: $6 \times 6 = 36$ and $\frac{1}{2} \times 6 = 3$.
- Step 3: $36 + 3 = 39$. (Or: $\frac{13}{2} \times 6 = \frac{78}{2} = 39$.)
- Step 4: Area = **39 sq in**.

✓ Answer: **39 sq in**

56. Challenge. A rectangular yard is $8\frac{1}{2}$ ft by 6 ft. A shed that is 2.5 ft by 2 ft sits in the yard. How many square feet of yard are NOT covered by the shed?

- Step 1: Yard: $8\frac{1}{2} \times 6 = 48 + 3 = 51$ sq ft ($8 \times 6 = 48$ and $\frac{1}{2} \times 6 = 3$).
- Step 2: Shed: $2.5 \times 2 = 5$ sq ft ($25 \times 2 = 50$, one decimal place $\rightarrow 5.0$).
- Step 3: Uncovered area (enclose and subtract): $51 - 5 = 46$.
- Step 4: Notice $8\frac{1}{2}$ and 2.5 use different notation, but the multiplication works the same way.
- Step 5: Uncovered area = **46 sq ft**.

✓ Answer: **46 sq ft**